Jacomo Corrieri

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EDUCATION

University of Florida, *Gainesville, FL* — *Expected May 2027*
Master of Science (M.S.) in Computer Science
GPA: 3.83/4.0
Relevant coursework: Machine Learning, Machine Learning Engineering, Computer Vision

University of North Florida, *Jacksonville, FL* — *May 2025*
Bachelor of Science (B.S.) in Computer Science, Minors in Mathematics and International Business
GPA: 3.99/4.0
Relevant coursework: Machine Learning with Graphs, Research in Graph Networks, Vector Calculus

Sungkyunkwan University, *Seoul, South Korea* — *Feb 2024 – Jun 2024*
Exchange Program, Data Science
- Completed coursework in exploratory data analysis (EDA) and AI applications.

WORK EXPERIENCE

Florida Blue — *May 2026 – Present*
Information Technology Intern
- Developed a full-stack interface using Streamlit to support multi-team data migration and validation efforts.
- Created a robust backend REST API using FastAPI, Pydantic, and SQLAlchemy, delivering sub 100ms response times.
- Maintained application quality through integration and unit testing with Pytest, and linting and formatting with Ruff.
- Translated business requirements into Agile user stories, aligning with company design standards.

UF Information Technology — *Jan 2026 – Aug 2026*
Research Computing Support Intern
- Built a Python CLI and API enabling staff analysis of 800M+ rows of software usage data on HiPerGator HPC.
- Containerized and deployed the API and PostgreSQL database to production using Podman Quadlet.
- Optimized query performance using nightly precomputed tables, reducing latency from minutes to milliseconds.

PROJECTS

Multi-Agent Area Coverage via Deep Reinforcement Learning
- Trained multi-agent RL systems (CNN, DQN, PPO) using PyTorch and Ray RLlib for large-scale area coverage.
- Achieved ~99% coverage and connectivity with 20–50 robots, 15% higher average coverage than prior work.
- Built a modular experimentation framework enabling rapid iteration on reward functions and hyperparameters.

MoTorch Neural Network Library
- Architected a PyTorch-inspired deep learning framework with neural network modules, optimizers, and loss functions.
- Engineered a reverse-mode autodiff engine using a dynamic computational graph for backpropagation.
- Leveraged GitHub Actions to facilitate linting, formatting, and testing in a CI pipeline and uv for fast package builds.

DQN Pathfinding Paper Implementation
- Implemented a deep reinforcement learning paper using PyTorch, Gymnasium, and Stable-Baselines3.
- Derived CNN-based DQN architecture from the paper and created unique testing and training environments.
- Logged performance and relevant metrics using TensorBoard, comparing results against an A* agent.

SKILLS

- Languages: Python, Go, C, Java, HTML/CSS, SQL, TypeScript
- Frameworks: PyTorch, NumPy, Scikit-Learn, Gymnasium, Ray RLlib, FastAPI, Django/Ninja, Gin, Gorm
- Tools/Infra: Slurm, HPC, Docker/Podman (Quadlet), Linux, PostgreSQL, Git, Jira, GitHub Actions

AWARDS/CERTIFICATIONS

- Building Transformer-Based Natural Language Processing (NLP) Applications, *NVIDIA*
- 2nd place out of 50 teams, Fall 2024 UNF NestforAwhile Programming Challenge
- Unix Essential Training, *LinkedIn*